Linnæus University
Department of Computer Science and Media Technology
Amilcar Soares
Rafael M. Martins

Exam in Introduction to Machine Learning, 2DV516, 7.5 hp
June7, 2025, 8.00–13.00

The exam consists of 9 Questions, each with several subquestions. The maximum number of points for the exam is 52. In order to pass (grade E), you need 26 points. **All your answers should be written on the exam (multiple choice, short answers, and drawings). Loose sheets can be used to have your calculations and partial reasoning before giving your final choice or answer. Everything in the loose sheets will not be graded.**

1. k-Nearest Neighbors (kNN) (6 points)

1. **Which of the following best describes how KNN works for classification? (1 point)**
   Correct answer: (c) It finds the k closest training samples to a query point and uses majority voting.
   Justification: KNN is a lazy, instance-based learning algorithm that does not build an explicit model. For classification, it assigns the class based on the majority among the k nearest neighbors in the feature space.
   (a) It builds a decision tree to separate classes.
   (b) It computes the mean squared error over all features.
   (c) It finds the k closest training samples to a query point and uses majority voting.
   (d) It optimizes weights through gradient descent.
   (e) It clusters data points based on distance.

2. **What is the impact of setting k too low or too high in KNN? (2 points)**
   Correct answer: (b) Low k can lead to noisy predictions, high k may oversmooth decision boundaries.
   Justification: A low value of k (e.g., $k = 1$) means the classifier is highly sensitive to noise and outliers, leading to high variance and possibly overfitting. A very high k averages over many points, potentially including distant or irrelevant samples, resulting in smoother but less precise boundaries, which may increase bias. **Sample answer:** With a small k, KNN may overfit and be too sensitive to individual data points. With a large k, the model might ignore local structure and misclassify points near class boundaries.
   (a) Low k leads to high bias, high k leads to overfitting.
   (b) Low k can lead to noisy predictions, high k may oversmooth decision boundaries.
   (c) Low k results in high variance, high k reduces training time.
   (d) k has no impact if the data is normalized.
   (e) Low k may oversmooth decision boundaries, high k can lead to noisy predictions.

   Justify your answer below. (You cannot use more than the specified lines.)

3. **In a highly imbalanced dataset (e.g., 90% Class A, 10% Class B), which of the following is a reasonable strategy to help KNN deal with the imbalance? (2 points)**
   Correct answer: (e) Use distance-weighted voting.
   Justification: In imbalanced datasets, the majority class can dominate the k neighbors, leading to biased predictions. Distance-weighted voting reduces the influence of more distant neighborsâif a minority class sample is closer than the others, it can have a larger impact even if outnumbered.
   (a) Normalize the dataset
   (b) Use Euclidean distance instead of Manhattan

(c) Reduce the number of neighbors k
(d) Increase the number of neighbors k
(e) Use distance-weighted voting

Justify your answer below. (You cannot use more than the specified lines.)

4. **Which of the following is NOT a potential issue with using KNN in real-world applications? (1 point) Correct answer: (a)** It easily handles categorical variables without preprocessing.

   Justification: KNN is distance-based, and categorical variables cannot be used directly without encoding (e.g., one-hot or ordinal encoding). Thus, saying it "easily handles" them is incorrect. All other choices correctly identify limitations: KNN does not scale well, is computationally expensive at inference time, is sensitive to irrelevant features, and becomes less interpretable as the number of features increases.

   (a) It easily handles categorical variables without preprocessing.
   (b) It requires the model to be retrained for every prediction.
   (c) It does not scale well with large datasets.
   (d) It is sensitive to irrelevant features.
   (e) KNN becomes less interpretable as the number of features increases.

2. Linear Regression and Bias-variance tradeoff. (6 points)

1. **You apply linear regression to a dataset and get the following results (1 point):**
   - Model A (Simple): Training error = 4.5, Validation error = 5.0
   - Model B (Complex): Training error = 1.2, Validation error = 8.0

   Which model is likely overfitting?

   (a) Model A
   (b) Model B
   (c) Both models
   (d) Neither model
   (e) Cannot determine without test data

   Justify your answer. (You cannot use more than the specified lines.)

   Correct answer: (b)

   Justification (sample): Model B fits the training data well but performs poorly on validation, which suggests high variance and overfitting.

2. **Which type of error is most directly affected by model complexity? (2 points)**

   (a) Bias
   (b) Variance
   (c) Noise
   (d) Overfitting
   (e) Sample size

   Briefly explain why. (You cannot use more than the specified lines.)

   Correct answer: (b) Justification sample: As model complexity increases, variance typically increases because the model becomes more sensitive to the training data.

3. **Match each statement to the most appropriate concept. Write the correct letter (A - D) next to each number. (3 points)**

   A. High bias
   B. High variance

C. Low error

D. Neither bias nor variance

(1) Model underfits the data. __________

(2) Model's predictions vary drastically with different training samples. __________

(3) Model generalizes well to unseen data. __________

(4) Model shows consistently poor performance on both training and validation. __________

(5) Model shows large difference between training and validation errors. __________

(6) Model fits both training and validation data with low error. __________

(7) Model performs inconsistently across different test sets. __________

(8) Model ignores regularization penalties. __________

Sample correct answers:

- (1) A
- (2) B
- (3) C
- (4) A
- (5) B
- (6) C
- (7) B
- (8) D

3. Logistic Regression. (6 points)

1. **A logistic regression spam filter shows 95% accuracy, but most actual spam emails are missed. Which metric would be more informative for this case? (2 points)**

   (a) Recall
   (b) Precision
   (c) F1-score
   (d) ROC AUC
   (e) Confusion matrix

   Briefly justify your answer. (You cannot use more than the specified lines.)

   Correct answer: (a) Justification: Recall focuses on how many actual positive instances (spam emails) are correctly identified. Low recall in this case suggests false negatives.

2. **A logistic regression model outputs a score of 0.68 for a customer buying a product. You want to automate sending promotional emails only to likely buyers. What is a reasonable strategy for setting a threshold? (2 points)**

   (a) Use 0.5 as the universal threshold
   (b) Choose the threshold to maximize model training accuracy
   (c) Lower the threshold until 100% of users get emails
   (d) Use the mean of all predicted probabilities as the threshold
   (e) Choose a threshold based on business trade-offs (e.g., precision vs. recall)

   Briefly justify your choice. (You cannot use more than the specified lines)

   Correct answer: (e) Justification (sample): The threshold should be selected to reflect business goals; for example, if false positives are costly, you might prefer a higher threshold to improve precision.

3. **Why might logistic regression perform poorly in highly imbalanced binary classification tasks? (2 points)**

   (a) It assigns random probabilities to all samples
   (b) It ignores the cost of false negatives
   (c) It cannot output probabilities
   (d) It optimizes overall accuracy, which favors the majority class
   (e) It always predicts the minority class

   Briefly justify your choice. (You cannot use more than the specified lines)

   Correct answer: (d) Justification: Logistic regression minimizes a loss function that doesn't inherently adjust for imbalance, leading to majority-class bias.

4. Model selection and regularization. (6 points)

1. **Match each regularization or model selection concept (A - H) to its corresponding description (1 - 8). (3 points)**

   A. L1 Regularization
   B. L2 Regularization
   C. Hyperparameter tuning
   D. Cross-validation
   E. Early stopping
   F. Feature selection
   G. Bias
   H. Variance

   (1) Reduces weights but does not eliminate them entirely. ________
   (2) May cause underfitting when too strong. ________
   (3) Process to choose values like regularization strength. ________
   (4) Stops training when validation error starts increasing. ________
   (5) Leads to different models with small data changes. ________
   (6) Can shrink coefficients to exactly zero. ________
   (7) A strategy to estimate generalization performance. ________
   (8) Removes inputs irrelevant to target variable. ________

   Correct answers: (1) B, (2) G, (3) C, (4) E, (5) H, (6) A, (7) D, (8) F

2. **The table below shows training and validation performance for four models with increasing regularization. Answer all the subquestions below. (3 points)**

Model	Training Accuracy (%)	Validation Accuracy (%)	Num. Non-Zero Weights
A	99.0	87.0	35
B	96.5	89.8	20
C	94.0	91.2	15
D	91.0	85.0	7

(a) Which model shows signs of overfitting? Why?

____

(b) Which model would you deploy? Justify your answer.

____

____

(c) Based on the trend in the number of non-zero weights, what effect is increasing regularization having on these models? Choose the best interpretation:

(a) Regularization is increasing both training and validation accuracy.
(b) Regularization is shrinking weights and simplifying the model.
(c) Regularization is making the models more complex.
(d) Regularization has no effect on model sparsity.
(e) Regularization is increasing overfitting.

Expected answers: (a) Model A (b) Model C (best validation accuracy + simpler model than A/B) (c) Answer is b

5. Neural Networks. (6 points)

1. **Which of the following best explains why neural networks have recently become highly successful in solving complex tasks? (1 point)**
   - (a) Because they now use decision trees to enhance predictions
   - (b) Because the perceptron was proven to be sufficient for all problems
   - (c) Because neural networks no longer require activation functions
   - (d) Because of advances in computational power, large datasets, and improved training algorithms
   - (e) Because neural networks were recently discovered

   Correct answer: (d)

2. **Which of the following statements best describes the role of the activation function in a perceptron or neuron? (1 point)**
   - (a) It replaces the need for weights and biases
   - (b) It maps the input to a feature space using decision trees
   - (c) It introduces non-linearity into the model, allowing the network to learn complex functions
   - (d) It prevents overfitting by skipping layers
   - (e) It ensures output always stays within the 0 - 1 range

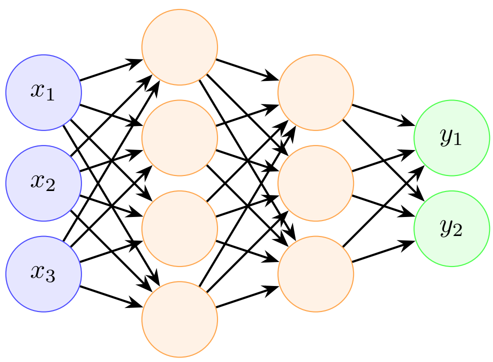
   Correct answer: (c)

3. **What is the main role of the backpropagation algorithm in training a neural network? (2 points**
   - (a) To create more hidden layers during training
   - (b) To minimize the loss function by computing weight gradients
   - (c) To shuffle the training data and prevent overfitting
   - (d) To convert predictions into probabilities
   - (e) To initialize the weights with small random values

   Briefly explain your answer

   Correct answer: (b) Expected justification: Backpropagation computes how much each weight contributes to the overall error by applying the chain rule, enabling the model to update weights to minimize the loss.

4. **The Figure below represents a feedforward neural network that uses the sigmoid function as the activation function. (2 points)**



   (a) Describe the network in details. For example, the number of: input, output, neurons. How many and what type of layers does this network have?

(b) What makes this model more powerful than a simple perceptron?

Expected answers: (a) 4 layers (1 input + 2 hidden + 1 output) (b) Multiple layers with non-linear activations allow the model to learn more complex, hierarchical representations.

6. Decision Trees and Ensemble Methods. (6 points)

1. **What is the goal of a decision treeâs splitting criterion at each internal node? (1 point)**
   (a) To find the split that creates the largest possible leaf nodes
   (b) To randomly divide the dataset for ensemble learning
   (c) To maximize entropy in the child nodes
   (d) To choose a split that improves class separation (e.g., by reducing impurity)
   (e) To minimize the number of features used

   Correct answer: (d)

2. **Match each concept (A - H) with its corresponding description (1 - 8). (3 points)**
   A. Entropy
   B. Information gain
   C. Gini index
   D. Bagging
   E. Random Forest
   F. Boosting
   G. Overfitting
   H. Pruning

   (1) A metric used to measure node impurity in classification tasks. ____________
   (2) Combining weak learners sequentially, focusing on hard examples. ____________
   (3) Measures reduction in impurity from a split. ____________
   (4) The tendency of deep trees to memorize training data. ____________
   (5) Reduces complexity by removing less important branches. ____________
   (6) The average of multiple trees trained on bootstrapped samples. ____________
   (7) Ensemble that introduces feature randomness during splitting. ____________
   (8) A measure of uncertainty or disorder in a dataset. ____________

   Answer key: (1) C, (2) F, (3) B, (4) G, (5) H, (6) D, (7) E, (8) A

3. **Which of the following best describes why a Random Forest helps reduce overfitting compared to a single decision tree? (2 points)**
   (a) It eliminates deep branches from all trees
   (b) It averages predictions from multiple uncorrelated trees, reducing variance
   (c) It uses more features per tree to improve accuracy
   (d) It trains on smaller datasets to simplify learning
   (e) It applies pruning aggressively on each tree

   Briefly explain your answer:

   ____

   ____

   Correct answer: (b) Justification: Random Forest reduces overfitting by averaging predictions of diverse trees trained on bootstrapped data and random subsets of features, reducing model variance.

7. Support Vector Machines. (6 points)

1. **Which of the following most accurately describes how the SVM regularization parameter influences bias and variance? (1 point)**

   (a) A high value increases bias and reduces variance
   (b) A low value increases both bias and variance
   (c) A high value reduces bias and increases variance
   (d) A low value makes the margin tighter
   (e) A high value enforces feature selection

   Correct answer: (c)

2. **Which of the following is a disadvantage of using kernel-based SVMs in real-world applications? (1 point)**

   (a) They do not require labeled data
   (b) They always require feature scaling
   (c) They are highly interpretable and easy to visualize
   (d) They are difficult to interpret and computationally expensive
   (e) They only work with decision trees

   Correct answer: (d)

3. **In your own words, explain the role of support vectors in determining the SVM decision boundary. (2 points)**

   Expected response: Support vectors are the critical training examples that lie closest to the margin. They directly determine the position of the decision boundary â all other data points have no influence.

4. **You trained four SVM models with different kernel settings and hyperparameters. Their performance and characteristics are shown below: (2 points)**

Model	Kernel	Training Acc. (%)	Validation Acc. (%)	Support Vectors	Training Time (s)
A	Polynomial (degree 5)	98.5	80.0	210	8.2
B	Gaussian kernel	95.0	87.5	190	5.5
C	Linear	93.0	91.0	130	2.1
D	Gaussian kernel (high gamma)	99.8	78.0	300	12.3

(a) Which model shows signs of overfitting or underfitting? Briefly explain for each if applicable.

Model A â Likely Overfitting

Very high training accuracy (98.5%) but low validation accuracy (80.0%)

High number of support vectors suggests a complex, possibly too-flexible decision boundary

Polynomial kernel (degree 5) often leads to overfitting with high-degree terms

Model B â Generalizes Well

Balanced training and validation accuracy (95.0% / 87.5%)

Reasonable number of support vectors (190)

Gaussian kernel handles non-linear relationships effectively without overfitting

Model C â Possibly Slight Underfitting, but Strong Generalization

Training accuracy is the lowest (93.0%), suggesting it may underfit slightly

However, validation accuracy is the highest (91.0%), with the fewest support vectors (130) and fastest training time

Simpler linear decision boundary may be sufficient if data is mostly linearly separable

Model D â Clear Overfitting

Extremely high training accuracy (99.8%) but very poor validation accuracy (78.0%)

Very high number of support vectors (300)

High gamma in the Gaussian kernel leads to decision boundaries that fit noise

(b) If your goal is to deploy a robust, interpretable, and efficient model in a real-time system, which model would you choose? Justify your decision using the trade-offs in accuracy, support vectors, and training time.

Recommended model: Model C (Linear kernel)

Justification: Highest validation accuracy (91.0%) â best generalization to unseen data Lowest number of support vectors (130) â fastest prediction time (important in real-time systems) Fastest training time (2.1 seconds) High interpretability -> decision boundary is linear and explainable While Model B is also a strong choice, Model C performs better overall for a system that values speed, simplicity, and explainability

8. Clustering. (5 points)

In an alternate reality, nobody won World War II. After 80 years of non-stop fighting, the remaining 15 countries (see Table 1) in Europe decided to simply split the continent in 3 regions and try to live in peace again. They chose to use a clustering algorithm for that, to try and get neutral results.

Distance	Ams	Ath	Ber	Brn	Bru	Cop	Edi	Lis	Lon	Lux	Mad	Par	Pra	Rom	Vie
Amsterdam	0	3082	649	875	209	904	1180	2300	494	371	1782	515	973	1835	1196
Athens	3082	0	2552	2627	3021	3414	3768	4578	3099	2744	3940	3140	2198	2551	1886
Berlin	649	2552	0	986	782	743	1727	3165	1059	767	2527	1094	354	1573	666
Bern	875	2627	986	0	655	1392	1643	2179	975	429	1541	556	766	897	907
Brussels	209	3021	782	655	0	1035	996	2080	328	233	1562	294	911	1615	1134
Copenhagen	904	3414	743	1392	1035	0	1864	3115	1196	1106	2597	1329	1033	2352	1345
Edinburgh	1180	3768	1727	1643	996	1864	0	2879	656	1206	2372	1082	1872	2467	2098
Lisbon	2300	4578	3165	2179	2080	3115	2879	0	2210	2165	638	1786	2945	2737	3255
London	494	3099	1059	975	328	1196	656	2210	0	538	1704	414	1204	1799	1233
Luxembourg	371	2744	767	429	233	1106	1206	2165	538	0	1647	379	746	1400	885
Madrid	1782	3940	2527	1541	1562	2597	2372	638	1704	1647	0	1268	2307	2099	2617
Paris	515	3140	1094	556	294	1329	1082	1786	414	379	1268	0	1094	1531	1285
Prague	973	2198	354	766	911	1033	1872	2945	1204	746	2307	1094	0	1370	312
Rome	1835	2551	1573	897	1615	2352	2467	2737	1799	1400	2099	1531	1370	0	1168
Vienna	1196	1886	666	907	1134	1345	2098	3255	1233	885	2617	1285	312	1168	0

Table 1: Driving distances (km) between the 15 remaining capitals of Europe. All the roads are two ways, so the distances are symmetric. Source: `https://www.engineeringtoolbox.com/`

(a) Computers have not advanced as much in this parallel universe, so people do machine learning by hand instead. Perform an agglomerative hierarchical clustering on the capitals using single linkage, and draw the corresponding dendrogram. **Note:** draw it horizontally, and the scale does not have to be accurate. (3p)

(b) Split your dendrogram in 3 clusters. Is the new world order a balanced scenario? Why? (1p)

(c) Which city is the biggest outlier in the dataset? In this context, an outlier is a city that is far away from most (or all) other cities. (1p)

9. Dimensionality Reduction. (5 points)

Figure 1 is an example of the results of applying two different dimensionality reduction methods (PCA and Sammon mapping) to the same dataset.

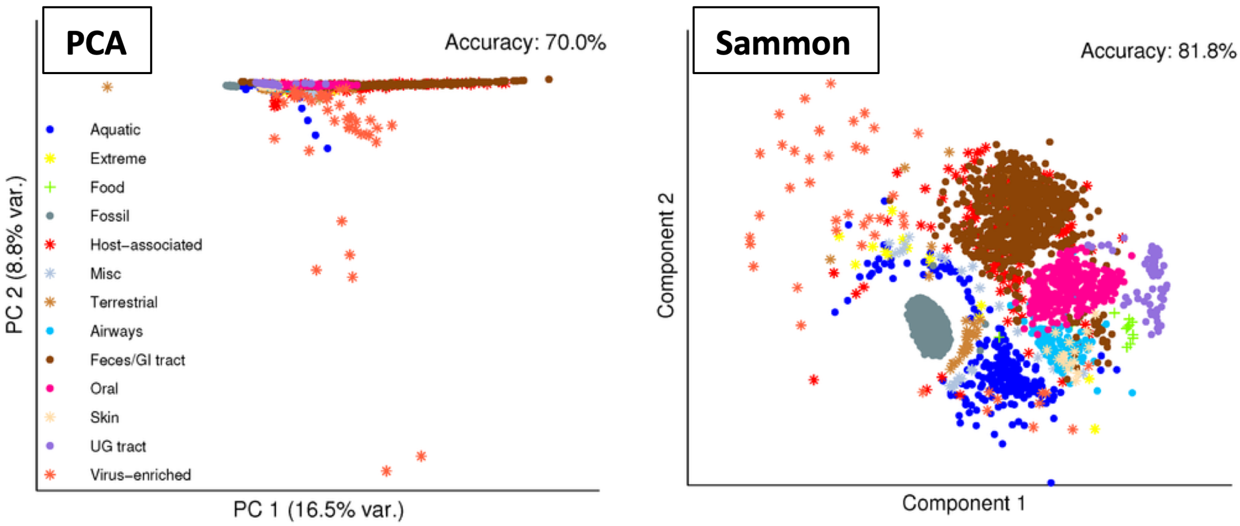


Figure 1: PCA vs. Sammon mapping. Source: `https://www.mdpi.com/1422-0067/15/7/12364`

The questions below are regarding the methods in general, not the figure specifically (you should use the figure as an example only).

(a) What are the main differences between these two methods, which cause them to yield so different results? (Tip: think about the optimization process involved in both cases) (2p)

(b) What is the unique characteristic of Sammon mapping that might make it more attractive than PCA in some occasions? (1p)

(c) Why would an analyst choose a complex non-linear DR (such as Sammon or t-SNE) over a simple linear one (such as PCA)? (1p)

Good Luck!